MATH 142: Exam 2 **Name:** ____________________

Spring —$_1$ 2026

03/19/2026

75 Minutes

Write your name on the appropriate line on the exam cover sheet. This exam contains 8 pages (including this cover page) and 7 questions. Check that you have every page of the exam. Answer the questions in the spaces provided on the question sheets. Be sure to answer every part of each question and show all your work and fully justify all your responses. If you run out of room for an answer, continue on the back of the page — being sure to indicate the problem number.

Question	Points	Score
1	16	
2	8	
3	8	
4	16	
5	16	
6	18	
7	18	
Total:	100	

1. (16 points) For each of the following series, determine whether the series converges or diverges. If the series converges, find the *exact* value of the sum—if possible.

   (a) $\displaystyle\sum_{n=5}^{\infty} \cos\left(\frac{1}{n^2}\right)$

   (b) $\displaystyle\sum_{n=1}^{\infty} \left(\sqrt{n+1} - \sqrt{n}\right)$

2. (8 points) Consider the series given below.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}} = 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \cdots$$

(a) Use an appropriate series test to show the series converges. Be sure to fully justify any necessary criterion.

(b) The sum of the first 63 terms, S_{63}, is approximately 0.667643. What is a maximum possible error bound for this estimation for the sum of the given series? Justify your error bound.

(c) What is a minimal number of terms one must add to approximate the value of the series to two decimal values, i.e. with an error less than $0.01 = \frac{1}{100}$? Justify your reasoning.

3. (8 points) Determine whether the following series converges or diverges. If the series converges, find the *exact* value of the sum—if possible.

$$\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{5^n}$$

4. (16 points) For each of the following series, determine whether the series converges or diverges. If the series converges, find the *exact* value of the sum—if possible.

(a) $\displaystyle\sum_{n=0}^{\infty} \frac{n^{10}\, 7^n}{n!}$

(b) $\displaystyle\sum_{n=2}^{\infty} \left(\frac{1-n^2}{3n^2+5}\right)^n$

5. (16 points) For each of the following series, determine whether the series converges or diverges. If the series converges, find the *exact* value of the sum—if possible.

   (a) $\displaystyle\sum_{n=10}^{\infty} \frac{\sqrt{n}}{n-9}$

   (b) $\displaystyle\sum_{n=1}^{\infty} \sin\left(\frac{1}{n^6}\right)$

6. (18 points) Determine whether the following series converges or diverges. If the series converges, determine whether the series converges conditionally or converges absolutely.

$$\sum_{n=3}^{\infty}(-1)^n \frac{n^{13}-1}{n^{14}+8}$$

7. (18 points) Use the Integral Test to determine whether the following series converges or diverges. If the series converges, find the *exact* value of the sum—if possible.

$$\sum_{n=2}^{\infty} \frac{1}{n \ln n}$$